

Changes in Gene Expression Level in Short Term Spheroid Culture of High-Grade Serous Ovarian Cancer (HGSOC) Patients

Abstract

High-Grade Serous Ovarian Cancer (HGSOC) is the most common subtype of epithelial ovarian cancer. With high genomic instability, somatic mutations in the tumour suppressor TP53 are found in most HGSOC samples. The disease is often diagnosed at advanced stages, leading to high mortality. The use of personalized medicine, including patient-derived multicellular spheroids that mimic natural tumour growth, is being studied to develop more effective treatments, especially for patients with aggressive or late-stage cancers.

To better understand gene expression patterns in short-term HGSOC spheroid cultures, differential gene expression analysis was performed on transcriptomics data collected from four patients at Day 0 and Day 6. All genes with statistical significance were annotated with up-regulation, down-regulation, and N/A. Top differentially expressed genes were identified and further studied. After visualization of the results, significant differences were observed between Day 0 and Day 6 spheroid cultures and between patient samples, with patient-specific variation showing a stronger effect than culture conditions. Literature analysis of the most differentially expressed genes identified several pathways associated with the observed transcriptional changes.

Introduction

Nearly 90% all ovarian cancers are epithelial.¹ High-Grade Serous Ovarian Cancer (HGSOC) is the most common subtype of the more aggressive type II epithelial ovarian cancer, accounting for 70% of all epithelial ovarian cancer.¹ Genetically, HGSOC can be characterized by its high genomic instability, especially in tumour suppressor TP53 as more than 96% of HGSOC samples collected contained some form of somatic TP53 mutation.² Because early stage symptoms are vague, the disease tends to get diagnosed after metastasis has occurred.^{1,3} Due to its later diagnosis, HGSOC is responsible for 73.9% of epithelial ovarian cancer deaths.¹ Effective treatment is necessary to reduce the mortality of HGSOC.

To increase treatment efficiency, reduce side effects, and improve patient survival, personalized medicine is often used for cancer patients. Patient-derived organoids (PDOs) can be used to develop genomic models to better understand the patient's disease or to test drug responses *ex vivo* to identify the most effective treatment for the patient.^{4,5}

Spheroids

- Multicellular aggregates that form in the abdominal cavity
- Complicates cancer treatment by conferring resistance to common treatment and promoting recurrence
- Also considered for use as accurate models of *in vivo*, patient-specific tumours⁶
- Short-term of spheroid culture effects remain unclear

The dataset consists of eight individual files of culture transcriptomics data collected from four patients at Day 0 and Day 6. Each file includes the Ensembl gene ID, the common name of the gene, and the read count of the gene.

Research Focus

This project sought to examine how gene expression changed during the culturing of patient-derived HGSOC spheroids. The main questions that drove the analysis were:

1. Is there a significant change in gene expression between days 0 and 6 of spheroid culturing?
2. If significant gene expression changes are observed, are these changes significantly different between patients?
3. Do the most differentially expressed genes involve related pathways?

Methods

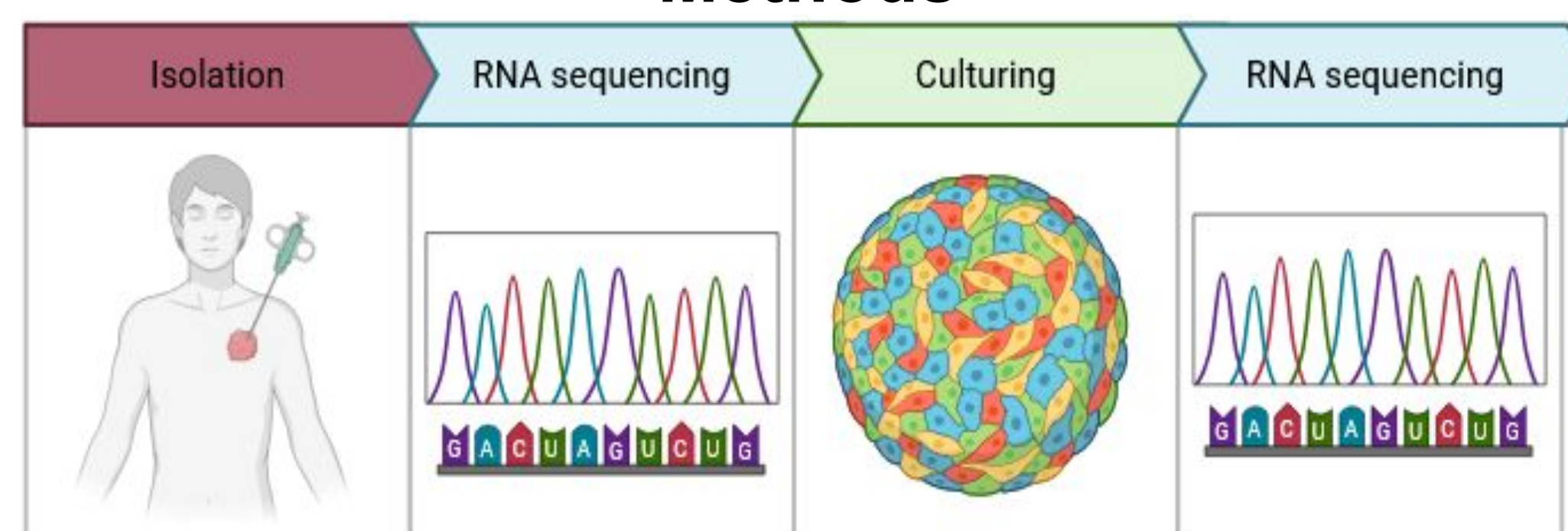


Figure 1) Representation of the data collection workflow. Patient spheroids were harvested at day 0, which was followed by RNA collection and sequencing. Spheroids were then cultured until day 6, at which point RNA was collected and sequenced again. A total of 4 patients were sampled (n=4)

Data Cleaning

- Rows containing counts not directly associated with a gene were removed
- Genes with 0 read counts on both days were also filtered out
- Genes with low counts overall (< 10) were also not counted

Statistical Analysis

Differential gene expression was performed using DESeq2, based on being from RNA seq.

- Expression was first compared by collection day to determine the effect of culturing on gene expression
- Expression was also compared by patient + day to control for patient-specific changes in gene expression

We accounted for false discovery rate (FDR) by using the Benjamini-Hochberg correction. Batch effect was investigated using statistical methods (DESeq) and visually with the PCA plot, corrected, and re-analyzed with corrections.

Results

The MA plot (Figure 2) shows a general trend where genes with lower base expression exhibit the highest changes in expression. This suggests the culturing process most strongly impacted genes not typically highly expressed in the tumours.

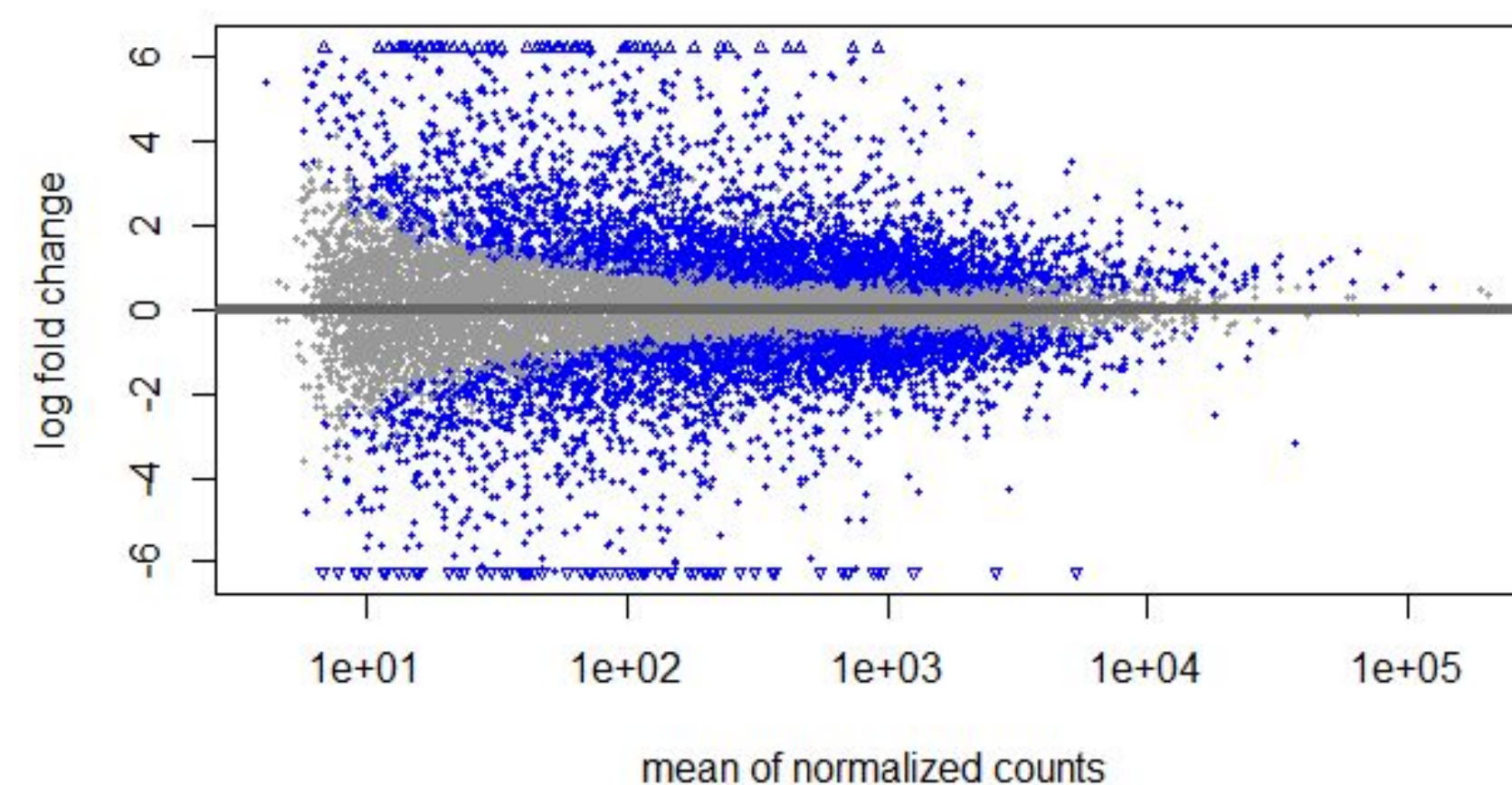


Figure 2) MA plot of differentially expressed genes with up- and down-regulated genes on top and bottom, blue indicating significant differentially expressed genes.

The volcano plot (Figure 3) builds on the MA plot by showing the significant differentially expressed genes (p-value >0.05, log2 change >1 or <-1).

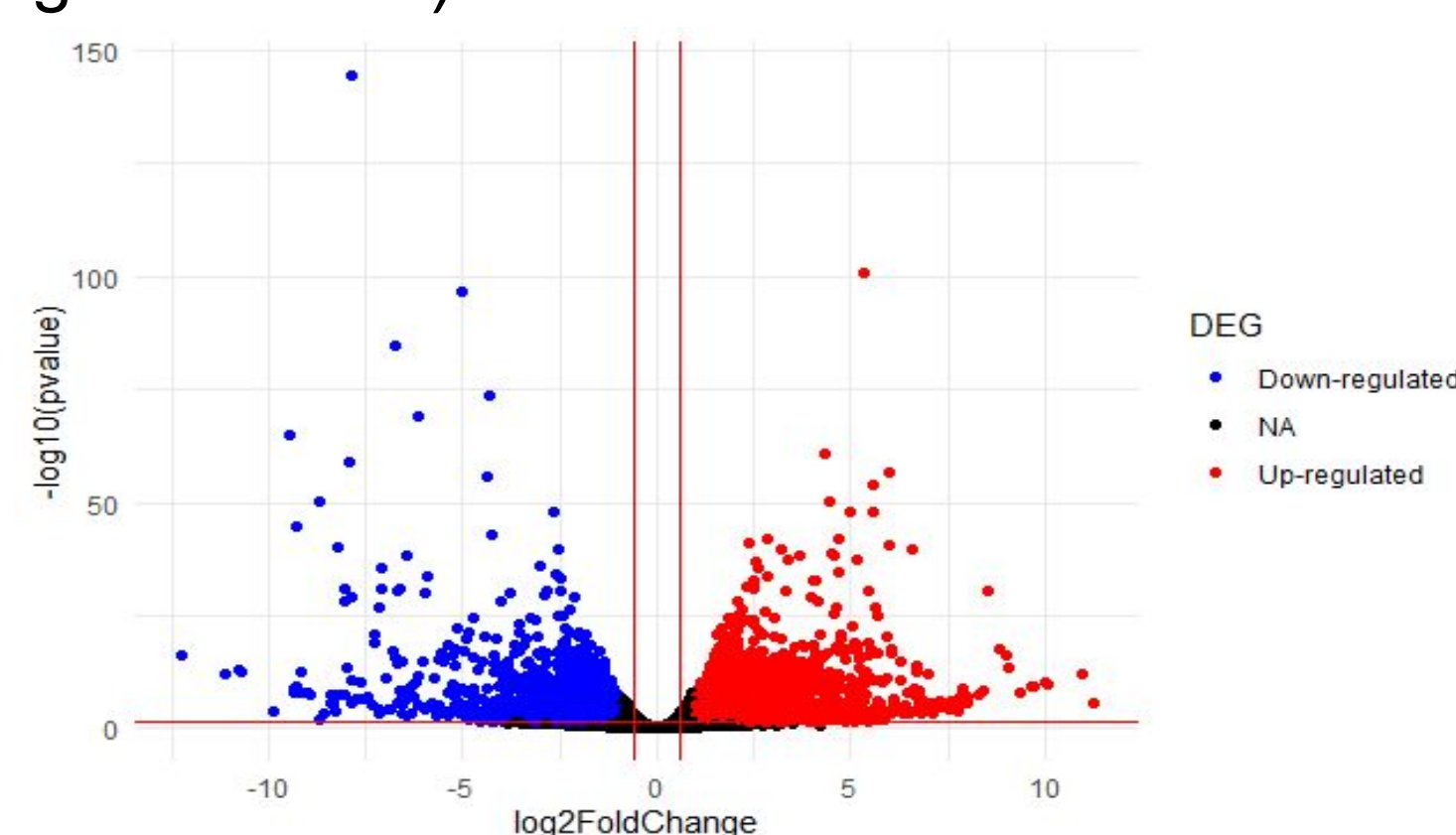


Figure 3) Volcano plot of differentially expressed genes with down-regulated genes in blue, up-regulated genes in red. Up/down regulation separated by +/- 0.05 p-value respectively, and greater than -log10(0.05)

The clustered heatmap (Figure 4) shows the top 10 differentially expressed genes. Gene expression patterns are more similar within patients across time points than between patients.

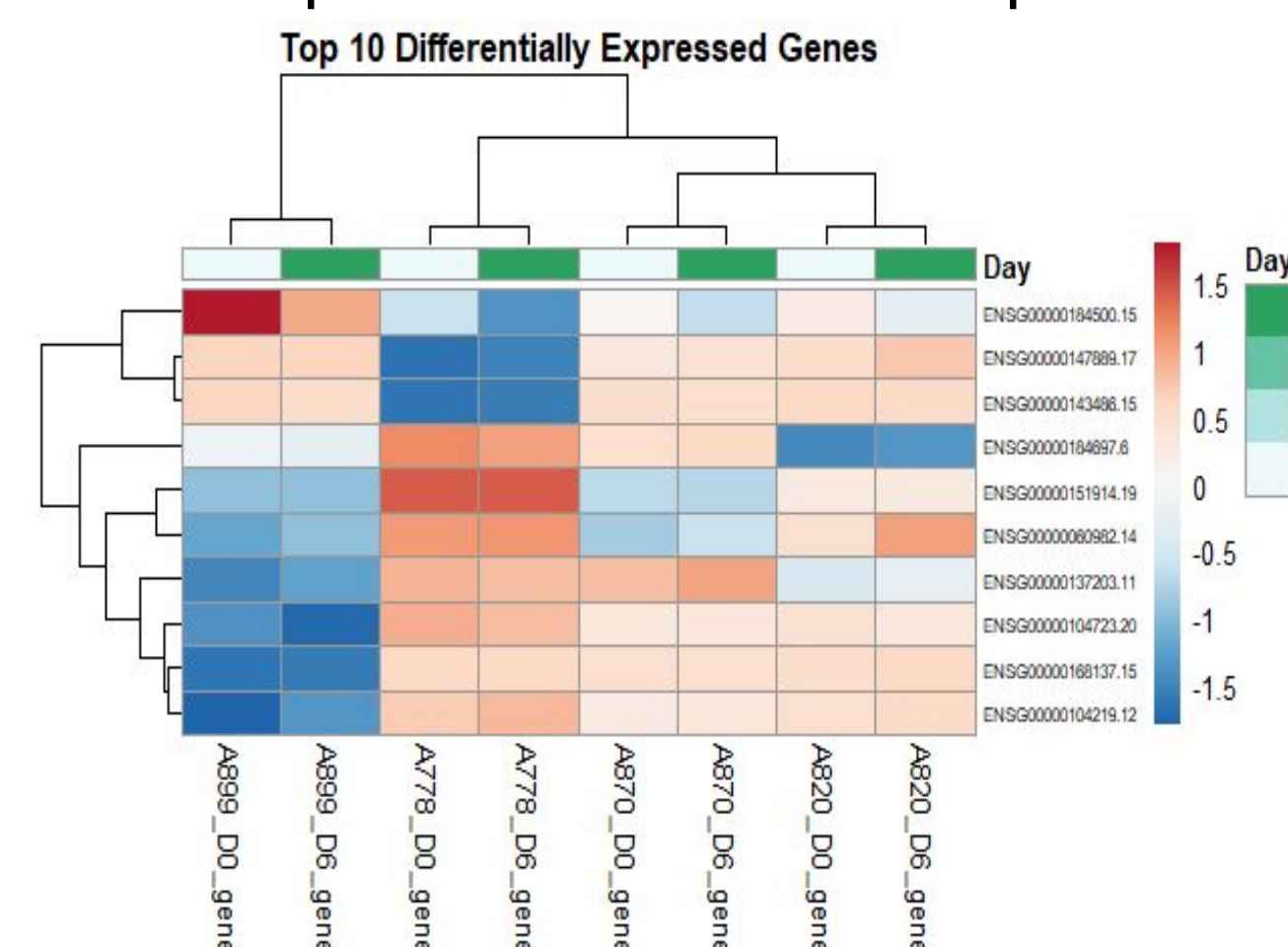


Figure 4) Clustered heatmap of the top 10 differentially expressed genes. Rows represent individual genes, columns represent samples at Day 0 (white) ,Day 6 (green).

Discussion

The RNA expression patterns of cultured HGSOC spheroids tended to group more closely by patient rather than time of collection, shown by both PCA plot and heatmap of the top differentially expressed genes.

- Suggests expression is most influenced by the tumour it originates from

The number and magnitude of up-regulated and down-regulated genes are comparable, however the changes in down-regulation show greater statistical significance.

ENSEMBL	SYMBOL	GENENAME	DEG
ENSG00000168137	SETD5	SET domain containing 5	Down-regulated
ENSG00000147889	CDKN2A	cyclin dependent kinase inhibitor 2A	Up-regulated
ENSG00000151914	DST	dystonin	Down-regulated
ENSG00000060982	BCAT1	branched chain amino acid transaminase 1	Down-regulated
ENSG00000184697	CLDN6	claudin 6	Down-regulated
ENSG00000137203	TFAP2A	transcription factor AP-2 alpha	Down-regulated
ENSG00000104723	TUSC3	tumor suppressor candidate 3	Down-regulated
ENSG00000184500	PROS1	protein S	Up-regulated
ENSG00000104219	ZDHHC2	zinc finger DHHC-type palmitoyltransferase 2	Down-regulated
ENSG00000143486	EIF2D	eukaryotic translation initiation factor 2D	Up-regulated

Table 1) Ensembl IDs, symbols, and names of top 10 differentially expressed genes. The table also indicates whether the gene is up-regulated or down-regulated (DEG).

Gene Functional Analysis

Several of the 10 most differentially expressed genes have roles relevant to either tumorigenesis or spheroid clustering:

- CDKN2A is a tumour suppressor that becomes up-regulated by day6, which correlates with spheroids being more stable growths
- CLDN6 and DST are both down-regulated genes involved in tight junctions and cytoskeletal formation

Conclusions

Differential gene expression p-value was significant (<0.01), meaning we reject our initial null hypotheses, meaning a significant change in gene expression was observed after 6 days of spheroid culturing, and gene expression patterns vary significantly between patient samples. An important note is that the differential expression between patient samples was more significant than culturing between Day 0 and Day 6.

Through a literature search, we were able to identify novel pathways that involved genes with the highest differential expression observed between patients and by day cultured.

Future Considerations

Address roadblocks to improve future studies:

- Increased number of sampling between initial culturing and final culture day
- Increased number of initial patient spheroid samples
- Culturing of normal ovarian cells from patient to determine a baseline/normal reference for expression analysis
- Better documentation of culture condition metadata

These findings were consistent with the findings found in other spheroid culture related studies involving gene expression.⁷Further confirming that HGSOC gene expression is more influenced by patient and conserved by extended periods of culturing.

References

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